

Aanchal Rathi

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Education

Vellore Institute of Technology — B.Tech, Computer Science (AI & ML Specialization) *Sep 2023 – Present*
CGPA: 8.98 / 10

DAV Public School — 12th Grade: 83.8% | 10th Grade: 94.1% *2019 – 2023*

Technical Skills

- **Languages:** Python, C++
- **Backend & APIs:** FastAPI, Flask, REST APIs, SQLAlchemy
- **Databases & DevOps:** PostgreSQL, MySQL, MongoDB Atlas, ChromaDB, AWS, Docker, Git/GitHub
- **Frontend:** HTML, CSS, JavaScript, React, Streamlit
- **AI / ML:** LLM APIs (Groq, Gemini, Llama), RAG, Scikit-learn, Pandas, NumPy

Projects

- **KhataFlow – AI-Assisted Ledger & Reconciliation Platform** *Jul 2026 – Present*
FastAPI, React, PostgreSQL, SQLAlchemy, Google Gemini API, Tesseract OCR
 - Built a document pipeline routing printed invoices through OCR and handwritten documents through a multimodal LLM, cutting API cost while handling messy real-world paperwork.
 - Designed a deterministic validation layer (GSTIN checksum, tax arithmetic) so extracted LLM output is never trusted directly before entering financial records.
 - Modeled an append-only ledger for a multi-party commission business, with FIFO payment matching and automatic commission-split calculation.
- **SEO Blog Agent – Multi-Service Content Generation Platform** *May 2026 – Jun 2026*
FastAPI, Streamlit, PostgreSQL, SQLAlchemy, ChromaDB, Docker, Render
 - Architected a two-service backend (FastAPI + Streamlit) with an async job-queue pattern, allowing 30–90s jobs to run without exceeding platform timeout limits.
 - Designed a PostgreSQL schema with connection pool recycling to fix long-running session stability issues on a serverless database.
 - Diagnosed and resolved production issues including memory crashes and connection pooling failures, shipping a containerized Docker deployment.
- **Youth News Hub – Personalized News Recommendation Platform** *Mar 2026 – May 2026*
Python, Flask, JavaScript, MongoDB Atlas, Firebase Authentication
 - Built a full-stack news aggregation platform integrating 2 news APIs with automatic fallback handling for resilient delivery.
 - Designed a recommendation engine to dynamically rank and personalize 15+ articles per request based on reading behavior.
 - Implemented Firebase Authentication and MongoDB Atlas schemas for scalable user preference and history storage.

Achievements & Certifications

- Microsoft Azure Data Fundamentals Certified.
- IBM SkillsBuild – Getting Started with AI, Certified.
- Machine Learning with Python – Coursera, Certified.
- Rank Holder, SOF NSO & IMO (2018 & 2019).

Leadership & Extracurriculars

- Core team member, University Dance Club – lead a team of 20-25 members through multiple college events including the university's flagship fest AdVITya. (2024–Present).
- Winner, Inter-College Women's Cricket Tournament (2024).